



Market-Making Workflow

User Guide — Curve Management, Revaluation & Risk

MKM Research Labs

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1 Introduction

The operator acts as a market-maker for Physical Risk Swaps, expressing views on flood risk by adjusting hazard curves and yield curves. When a curve is committed, every open trade is revalued and the portfolio P&L updates in real time.

This guide explains the market-making cycle: how curves are edited, how revaluation works, and how the trading desk tabs connect.

2 Market State

The **market state** is the complete set of inputs required to value every open PRS trade. It consists of:

Yield Curve A term structure of risk-free rates (1Y–6Y), used for discounting cashflows. Stored as continuously compounded annual rates.

Hazard Term Structure Per-gauge, per-trigger-level hazard rates across tenors (1Y–5Y). Drives the fair spread for each trade.

Risk-Free Rate A scalar fallback rate (default 4%) used when the yield curve is not available.

Base Rates Immutable reference hazard rates loaded from the gauge hazard curve file (`gaugehc.json`).

2.1 Persistence

The market state is persisted to `trading/market_state.json`. Each edit updates this file. The state survives server restarts and is captured in every EOD snapshot.

2.2 Reset

The operator can reset the market state to base curves at any time — either for a single gauge or for the entire portfolio. This reverts all adjustments and revalues trades at the original hazard rates.

3 Yield Curve Management

The yield curve panel (Market tab, “Yield” mode) displays the current term structure and allows point-by-point editing.

3.1 Editing

Each tenor point (1Y, 2Y, 3Y, 4Y, 5Y, 6Y) can be adjusted by:

- Dragging the chart control point up or down
- Typing a new rate in the input field

Edits are local until committed. A “dirty” indicator shows that unsaved changes exist.

3.2 Commit Workflow

Clicking **Commit** triggers:

1. **Snapshot before:** All open trades are valued at the current yield curve.
2. **Write curve:** The new yield curve is saved to `market_state.json`.
3. **Revalue all:** Every trade is repriced with the new discount factors. Fair spreads, MTMs, and deltas update.
4. **P&L impact:** The server calculates $\Delta\text{MTM} = \text{MTM}_{\text{after}} - \text{MTM}_{\text{before}}$ for each trade and returns a portfolio-level summary.
5. **Blotter refresh:** The UI switches to the Blotter tab, which now shows updated P&L columns.

3.3 Impact of Yield Curve Changes

A parallel upward shift in the yield curve:

- Increases discount factors, reducing the present value of future cashflows.
- Reduces the risky annuity A , which increases the fair spread (same protection PV divided by smaller annuity).
- Net effect on MTM depends on whether the trade is payer or receiver and the magnitude of the shift.

A typical 50 bp parallel shift moves fair spreads by 1–3 bps.

4 Hazard Term Structure

The hazard curve panel (Market tab, gauge mode) shows the current hazard term structure for a selected gauge and trigger level.

4.1 Three Trigger Levels

Each gauge has three independent hazard term structures:

Alert Lowest threshold — highest annual probability.

Warning Intermediate threshold.

Severe Highest threshold — lowest annual probability but largest potential loss.

PRS trades currently reference the severe trigger level, but the platform supports all three for research and scenario analysis.

4.2 Curve Shapes

The operator can express different views by reshaping the curve:

Parallel shift All tenors move by the same amount. Expresses a view that flood risk has uniformly increased or decreased.

Steepening Long-end rates rise relative to the short end. Suggests that near-term risk is stable but long-term risk is increasing (e.g. climate change effects).

Flattening Short-end rates rise relative to the long end. Suggests imminent risk but long-term improvement (e.g. flood defences under construction).

Inversion Short-end exceeds long-end. Unusual but could reflect a known near-term event.

4.3 Commit and Revaluation

The commit workflow for hazard curves is identical to the yield curve: snapshot before, write, revalue, P&L impact, blotter refresh. The difference is that hazard changes only affect trades referencing the edited gauge, whereas yield curve changes affect all trades.

5 Revaluation Cycle

The full revaluation cycle runs every time a curve is committed. The engine performs these steps for each open trade:

1. **Interpolate hazard rate** from the gauge's term structure at the trade's remaining tenor.
2. **Compute fair spread** via the CDS pricing formula (see Gauge PRS Pricing Guide).
3. **Compute risky annuity** for the remaining tenor.
4. **Compute MTM**: $(s_{\text{fair}} - s_{\text{trade}})/10,000 \times A \times N \times d$.
5. **Compute FS01**: bump hazard rate by 1 bp, recompute spread, derive sensitivity.

The revaluation is instantaneous for the typical portfolio size (10–50 trades).

6 Cross-Tab Navigation

The trading desk tabs are designed to flow naturally:

Market → **Blotter** After committing a curve change, the UI auto-switches to the Blotter to show updated P&L.

FS01 → **Blotter** Clicking a cell in the FS01 risk matrix (gauge × maturity) filters the Blotter to show only trades for that gauge at that tenor.

Blotter → **Stress** The Blotter's gauge filter carries forward as a hint when switching to the Stress tab, pre-selecting the gauge.

Market → History The “History” button shows a modal with the gauge’s hazard rates over all EOD dates, plotted as five tenor lines.

Market → P&L History The “PL Hist” button shows a modal with portfolio daily P&L and attribution.

7 Curve History

Every EOD snapshot captures the complete market state, including the yield curve and all hazard term structures. The Curves tab extracts this history and plots it as time series.

For a given gauge and trigger level, the chart shows five lines (1Y through 5Y hazard rates) evolving over all EOD dates. This reveals:

- Whether the operator has been systematically tightening or widening spreads.
- Seasonal patterns in hazard adjustment.
- Sudden jumps (e.g. after a flood event or a new storm catalogue generation).

8 Parameter Reference

Parameter	Default	Increase	Decrease
DEFAULT_YIELD_CURVE	3.5–4.5%	Lower PV of cash-flows, higher spreads	Higher PV, lower spreads
MAX_TOTAL_MOVE	varies	Allows larger curve adjustments	Constrains editing range
DAILY_HAZARD_VOL	varies	Wider daily hazard rate moves	Tighter daily moves
NUM_BUSINESS_DAYS	252	More trading days in annualisation	Fewer trading days
BUMP_1BP	0.0001	Larger FS01 bump	Finer sensitivity
RECOVERY	0.0	Reduces protection value	Full loss on event

Document History

Version	Date	Author	Description
1.0	10-Apr-2026	MKM Research Labs	Initial release